

Healthcare Appointment Optimization System

Business Insights & Recommendations

Executive Summary

Analysis of 20,000 appointments, 5,000 patients and 100 doctors shows that only 33.6% of appointments are completed, while 66.4% are cancelled or result in no-shows. This represents the biggest opportunity for improving operational efficiency and revenue.

Metric	Value	Insight
Appointments	20,000	Healthy operational dataset
Completed	6,724 (33.6%)	Low completion rate
Cancelled	6,583 (32.9%)	High revenue leakage
No Shows	6,693 (33.5%)	Major resource utilization issue
Average Wait	22.75 Days	Contributes to cancellations
Peak Hour	6:00 AM	Requires additional staffing

Key Insights

- Approximately two-thirds of appointments fail to convert into completed consultations.
- Double Booking is the largest controllable cancellation reason.
- Orthopedics, Pediatrics, ENT, General Medicine and Neurology have the highest no-show rates.
- Doctor workload is uneven across the organization.
- Patient demand peaks at 6 AM.
- Long waiting times reduce patient satisfaction.

Recommendations

1. Deploy automated appointment reminders.
2. Introduce real-time conflict detection to eliminate double bookings.
3. Optimize doctor allocation based on demand.
4. Reduce waiting times using waitlists and auto-fill scheduling.
5. Increase staffing during peak hours.
6. Monitor KPIs through executive dashboards.
7. Explore AI-based no-show prediction.

Expected Business Impact

- Reduce no-show rate by 30–40%.
- Increase completed appointments above 55%.
- Improve doctor utilization to 85–90%.
- Reduce waiting time below 14 days.
- Improve patient satisfaction and clinic profitability.

Roadmap

Phase 1 (0–3 Months): Reminders, scheduling validation.
Phase 2 (3–6 Months): Capacity optimization and waitlist automation.
Phase 3 (6–12 Months): AI scheduling and predictive analytics.

Conclusion

The analysis indicates that operational improvements can significantly enhance clinic performance without major infrastructure investments. Data-driven scheduling and resource optimization will improve efficiency, patient experience, and financial outcomes.